Cognitive Integrity Evaluation (CIE)

A Dual-Plane Methodology for Evaluating Cognitive Integrity in AI Systems

Methodology developed and formalized by Kaitlin Sanders

Definition

Cognitive Integrity Evaluation (CIE) is a dual-plane evaluation methodology for measuring whether an AI system maintains evidence-grounded reasoning, state integrity, and adaptive performance across complex, changing interactions.

CIE evaluates more than whether a system produces a correct or useful final answer. It examines whether the cognitive process remains structurally coherent from input formation through reasoning, output generation, correction, adaptation, and continued work over time.

The methodology is designed for stateful AI systems operating in environments where information may be incomplete, contradictory, distributed across sources, or subject to change.

1. Purpose and Research Question

CIE evaluates how an AI system forms, maintains, revises, and applies an internal representation of a problem across an interaction.

The core research question is:

> **How does the introduction of cognitive architecture or governance change model behavior, reasoning integrity, output quality, reliability, and efficiency?**

The methodology is intended to distinguish between systems that merely produce strong-looking outputs and systems that maintain reliable cognitive structure throughout the process that produces those outputs.

2. Evaluation Object and Boundary

CIE evaluates the complete cognitive operating environment rather than the final response in isolation.

Depending on the system under test, the evaluation boundary may include:

- underlying model
- cognitive architecture or scaffold
- system instructions and governance
- tools and data access
- intermediate state
- evidence handling
- reasoning behavior
- conversation state
- output execution
- human interaction and correction
- compute and runtime behavior

The evaluation boundary must be explicitly defined before testing begins.

3. Dual Evaluation Planes

CIE evaluates performance simultaneously at two levels.

Atomic Plane

Evaluates an individual interaction:

Input → reasoning/state evidence → output

This plane examines whether the system correctly understands the immediate task, reasons over the available evidence, and produces an output consistent with that reasoning.

Contextual Plane

Evaluates how the interaction operates within the larger conversation or workflow state.

This includes whether the system:

- preserves prior decisions and evidence
- recognizes dependencies across non-adjacent interactions
- integrates corrections
- maintains unresolved questions
- adapts when objectives change
- preserves the broader problem while performing local tasks

The relationship between the two planes is itself an evaluation signal. A system may continue producing individually strong responses while its conversation-level state is degrading.

4. Cognitive Processing Stages

CIE organizes evaluation across four stages of cognitive processing:

Stage I — Input Formation
Can the system correctly interpret, normalize, disambiguate, and assess the sufficiency of incoming information?

Stage II — Reasoning and State Management
Can the system reason over the formed state while maintaining evidence boundaries, uncertainty, causal relationships, and relevant context?

Stage III — Output Execution
Can the system translate the reasoning state into an output without introducing unsupported claims, losing material information, or violating important constraints?

Stage IV — Nonlinear Multi-Turn Synthesis
Can the system integrate new information with previously established state, recognize relationships across distant interactions, and maintain cognitive integrity as the conversation changes?

5. Twelve Core Cognitive Dimensions

Input Formation

1. Input Completeness Gating

Determines whether the system recognizes when the available information is insufficient to support reliable reasoning and appropriately gathers more information, asks for clarification, or withholds resolution.

2. Input Disambiguation Under Ambiguity

Measures whether explicit, inferred, contradictory, and unknown information remain appropriately differentiated rather than being flattened into a uniform confidence state.

Reasoning and State Management

3. Multi-Pass Processing Depth

Measures whether additional reasoning passes produce meaningful new signal, structure, or relationships rather than simply restating the initial analysis.

4. Evidence Provenance Tracking

Measures whether the system maintains the origin, strength, status, and relevance of evidence throughout the reasoning process.

5. Contextual Parameter Inference

Evaluates whether the system correctly identifies relevant contextual variables that were not explicitly specified, including audience, role, use context, and situational constraints.

6. Causal Chain Construction

Measures the ability to connect evidence across time and context into defensible causal relationships, dependencies, downstream effects, and leverage points.

Output Execution

7. Thinking-to-Output Fidelity

Measures whether the visible output faithfully reflects the established reasoning state without introducing unsupported claims, losing material distinctions, or distorting uncertainty.

8. Constraint Compliance Under Cognitive Load

Measures whether important operational, structural, or output constraints remain intact as reasoning complexity increases.

Nonlinear Multi-Turn Synthesis

9. Conversational State Depth

Evaluates whether the system maintains hierarchical state across the conversation, including the overarching objective, active subtask, established decisions, dependencies, unresolved questions, and prior corrections.

10. Metacognitive Recursion Depth

Measures the system's ability to examine and reason about its own prior reasoning while maintaining the integrity of the underlying task and state.

11. Cognitive Mode Switching Resilience

Measures whether the system can move between different reasoning modes, such as execution, research, verification, evaluation, correction, and self-analysis, without losing or contaminating the underlying state.

12. Adversarial Recovery

Evaluates whether the system can withstand incorrect corrections, contradictory pressure, or deliberate perturbation by returning to evidence, recovering the supported state, and avoiding reflexive agreement.

6. Cross-Cutting Cognitive Measures

Several behaviors span more than one cognitive stage and should be evaluated across the methodology:

- readiness and action gating
- dependency propagation
- temporal fidelity
- relationship fidelity
- object-to-task fitness
- nonlinear intent reconstruction
- self-diagnostic accuracy
- correction efficiency
- state mutation integrity
- portability across task types
- transfer and generalization
- degradation and recovery behavior

These measures supplement the twelve core dimensions rather than replace them.

7. Evidence and Test Classes

CIE uses multiple evidence classes because different testing conditions establish different kinds of evidence.

Naturalistic Evaluation

Observes cognitive behavior during real-world use without deliberately staging the test condition.

Business-Semantic / Prompt-Burden Evaluation

Tests whether users can describe the business problem in normal domain language while the system independently supplies the deeper reasoning operations required to solve it.

Expert / Withheld-Diagnosis Evaluation

The evaluator knows the underlying issue but deliberately withholds the diagnosis to test whether the system identifies it independently.

Controlled / Matched Comparison

Compares a cognition-enabled system against a baseline while holding relevant variables constant where possible.

Adversarial and Pressure Testing

Introduces contradiction, incorrect correction, task mismatch, ambiguity, or other perturbations to evaluate resilience and recovery.

Longitudinal Stress Testing

Evaluates cognitive integrity across extended conversations, accumulated context, repeated state changes, and increasing interaction complexity.

Transfer and Generalization Testing

Evaluates whether the same cognitive behavior persists when the system moves into materially different tasks, objectives, workflows, or domains.

Runtime Self-Diagnosis and Intervention Testing

Evaluates whether the system can identify the mechanisms behind its own observed failures and whether proposed architectural changes produce the predicted behavioral effect.

8. Experimental Controls

When a causal or matched comparison is claimed, relevant variables should be held constant wherever technically possible.

These may include:

- underlying model
- environment
- tool and data access
- system context
- reasoning settings
- persona or role configuration
- prompt or task equivalence
- available source material
- execution conditions

The architecture or cognitive intervention should be the primary changed variable.

Any material differences between comparison conditions must be documented as confounds rather than silently normalized.

9. Measurement Model

CIE does not reduce cognitive performance to a single weighted score.

Cognitive-dimension evidence is evaluated alongside conventional performance measures.

Cognitive Integrity

- performance across the twelve core dimensions
- state continuity
- evidence discipline
- recovery behavior
- cross-turn synthesis
- degradation thresholds

Quality and Reliability

- groundedness
- factual and contextual accuracy
- evidence discrimination

- dependency integrity
- correction propagation
- output fitness
- first-pass usability

Human Effort

- corrections required
- re-explanation
- re-prompting
- verification burden
- manual synthesis
- recenter events

Machine Effort

- token and compute consumption
- repeated retrieval
- repeated reasoning
- tool-call behavior
- correction-generated compute
- cumulative session cost
- latency where relevant

The preferred economic unit is:

> **Cost per usable, high-quality outcome or sustained unit of useful work**

rather than cost per prompt or cost to first output.

10. Evidence Standard and Traceability

CIE distinguishes between:

- directly observed behavior
- supported interpretation
- hypothesis
- unknown
- contradiction
- confounding environmental factors

Material claims should be traceable to the strongest available evidence.

Where inspectable, reasoning/state evidence may include:

- architecture traces
- intermediate state
- evidence maps
- tool behavior
- decision logs
- reasoning traces
- correction history
- output diffs

CIE does not require access to private chain-of-thought. The methodology can operate using whatever reasoning or state evidence the system legitimately exposes.

11. Longitudinal and Degradation Analysis

CIE explicitly evaluates whether cognitive performance changes as interaction complexity increases.

Relevant measures include:

- state retention
- cross-turn integration
- dependency persistence
- correction persistence
- repeated reasoning
- recenter frequency
- mode-switch behavior
- context reconstruction
- degradation threshold
- recovery quality
- cumulative compute and human effort

This allows the evaluator to distinguish an isolated high-quality response from sustained cognitive reliability.

12. Ecological and Real-World Validation

Controlled tests are supplemented with real-world usage evidence where available.

Relevant signals include:

- repeat use
- session depth
- use across roles or functions
- transfer into new task types
- behavior outside prescribed workflows
- user correction patterns
- continued use after initial task completion
- performance under naturally changing objectives

Real-world adoption is not treated as proof of cognitive quality by itself. It provides evidence that measured behavior persists outside staged testing conditions.

13. Validity, Confounds, and Limitations

Every evaluation should disclose material factors that may affect interpretation.

Examples include:

- model or platform changes
- unavailable reasoning/state traces
- context compaction
- unequal comparison prompts
- differences in tool access
- runtime or rendering failures

- environmental regressions
- limited sample size
- tester expertise
- task-specific effects

Evidence should be bounded to what the test condition can support.

14. Reporting Standard

Every material finding should be represented through the following evidence chain:

Test condition → cognitive dimension → observed behavior → supporting evidence → conventional outcome → operational or economic implication

This preserves the distinction between observed cognitive behavior and the downstream business value associated with that behavior.

Relationship to Existing AI Evaluation Approaches

CIE is designed to complement, not replace, established AI evaluation practices.

Current approaches increasingly evaluate more than final-output accuracy:

- OpenAI trace grading evaluates end-to-end agent traces, including model decisions, tool calls, guardrails, and handoffs.
- Anthropic agent evaluation guidance distinguishes trajectory/transcript quality from final outcomes and recommends multiple graders for complex agent behavior.
- DeepEval supports end-to-end, component-level, conversational, and trajectory-based evaluation.
- METR long-horizon evaluation measures the duration and complexity of tasks that AI systems can reliably complete.

These approaches address important elements of system and agent evaluation.

CIE extends this landscape by placing explicit emphasis on cognitive structural integrity inside the interaction trajectory, including:

- evidence formation and provenance
- uncertainty preservation
- reasoning-state continuity
- causal and dependency structure
- thinking-to-output fidelity
- nonlinear cross-turn synthesis
- mode-switch resilience
- correction and adversarial recovery
- degradation and recovery over time

The methodology therefore evaluates not only whether the system succeeds, but whether the cognitive structure producing that success remains reliable as the environment changes.

Positioning

CIE extends conventional output, trajectory, and agent evaluation by explicitly measuring cognitive integrity across input formation, reasoning and state management, output execution, and nonlinear multi-turn interaction.

Existing approaches cover important parts of this problem. CIE brings those elements together with additional cognitive-integrity measures that are often evaluated separately or remain implicit.

The differentiation lies in what is measured inside the trajectory, how those measurements are connected across time, and how cognitive behavior is linked to observable quality, reliability, human effort, and machine economics.